

Task 1: Scale Factor

To find the scale factor (S), we identify the maximum absolute value in the entire 4x4 weight matrix (W).

The element with the largest magnitude is -2.31.

Maximum absolute value: 2.31

Compute S: $2.31 / 127 = 0.018189$

Task 2: Quantize (Wq)

To compute the INT8 matrix, every element in the original matrix is divided by S, rounded to the nearest whole number, and clamped to the range [-128, 127].

47	-66	19	115
-4	50	-103	7
85	2	-24	-127
-10	57	42	30

Task 3: Dequantize (Wdeq)

To evaluate precision loss, the INT8 matrix is converted back to FP32 by multiplying every element in Wq by the scale factor S.

0.85488	-1.20047	0.34559	2.09173
-0.07275	0.90945	-1.87346	0.12732
1.54606	0.03637	-0.43653	-2.31000
-0.18189	1.03677	0.76393	0.54567

Task 4: Error Analysis

The absolute error for each element is calculated by finding $|W - W_{deq}|$.

Largest Error Element: The largest individual error occurs at the original 2.10 element (Row 1, Col 4). The dequantized value is 2.09173, yielding a maximum error of 0.00827.

Mean Absolute Error (MAE): Summing the absolute errors of all 16 elements yields approximately 0.06921. Dividing by 16 gives an MAE of 0.00432.

Task 5: Bad Scale Experiment (S = 0.01)

If we forcefully set $S = 0.01$, the quantization formula $\text{round}(W / 0.01)$ acts exactly like multiplying by 100.

Because an 8-bit integer can only hold values from -128 to 127, values exceeding this boundary are clamped:

2.10 becomes 210 → clamps to 127.

-1.88 becomes -188 → clamps to -128.

1.55 becomes 155 → clamps to 127.

-2.31 becomes -231 → clamps to -128.

Impact on Error: The massive errors on those clamped elements (e.g., $|2.10 - 1.27| = 0.83$) push the overall Mean Absolute Error (MAE) to 0.17125.

Conclusion: When the scale factor S is too small, large matrix values exceed the 8-bit limit and are severely clamped, causing massive precision loss and a significantly higher Mean Absolute Error.